

Maximum Likelihood Estimators (MLE)

Definition, Examples, and Key Properties

Md Irtiaz Kabir

Motivation — Bayes' Rule

We want to find the value of θ that makes θ most plausible given the data we observed. That is, we want to maximize

$$\mathbb{P}(\theta \mid \text{data}).$$

By **Bayes' rule**,

$$\mathbb{P}(\theta \mid \text{data}) = \frac{\mathbb{P}(\text{data} \mid \theta) \mathbb{P}(\theta)}{\mathbb{P}(\text{data})}.$$

So to maximize the left-hand side over θ , we can study how the right-hand side depends on θ .

Simplifying the Posterior

Suppose all values of θ are considered **equally likely** *a priori*. Then:

- $\mathbb{P}(\theta)$ is the **same** for every θ .
- $\mathbb{P}(\text{data})$ does not depend on θ at all — it is just a normalizing constant.

Key consequence

$$\mathbb{P}(\theta \mid \text{data}) \propto \mathbb{P}(\text{data} \mid \theta).$$

Maximizing the posterior is therefore the **same** as maximizing $\mathbb{P}(\text{data} \mid \theta)$.

The Likelihood Function

Let $X_1, \dots, X_n$ be IID with PDF $f(x; \theta)$. By independence,

$$\mathbb{P}(\text{data} \mid \theta) = \mathbb{P}(X_1, \dots, X_n \mid \theta) = \mathbb{P}(X_1 \mid \theta) \cdots \mathbb{P}(X_n \mid \theta) = \prod_{i=1}^n f(X_i \mid \theta).$$

Definition — Likelihood Function

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f(X_i; \theta).$$

The **log-likelihood function** is $\ell_n(\theta) = \log \mathcal{L}_n(\theta)$.

$\mathcal{L}_n(\theta)$ is just the joint density of the data, but viewed as a **function of θ** — it need not integrate to 1 over θ .

Definition — Maximum Likelihood Estimator

The **maximum likelihood estimator** (MLE), denoted $\hat{\theta}_n$, is the value of θ that maximizes $\mathcal{L}_n(\theta)$.

Since log is increasing, maximizing $\ell_n(\theta)$ gives the **same** maximizer as maximizing $\mathcal{L}_n(\theta)$ — it is often easier to work with the log-likelihood.

Remark

If we multiply $\mathcal{L}_n(\theta)$ by any positive constant c (not depending on θ), the MLE does not change. We may therefore drop such constants when convenient.

Example — Bernoulli

Suppose $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, with $f(x; p) = p^x(1 - p)^{1-x}$ for $x = 0, 1$.

Likelihood

$$\mathcal{L}_n(p) = \prod_{i=1}^n f(X_i; p) = \prod_{i=1}^n p^{X_i}(1 - p)^{1-X_i} = p^S(1 - p)^{n-S}, \quad S = \sum_i X_i.$$

Log-likelihood

$$\ell_n(p) = S \log p + (n - S) \log(1 - p).$$

Setting $\ell'_n(p) = 0$ and solving gives the MLE

$$\hat{p}_n = S/n.$$

Example — Normal (Part 1)

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, so $\theta = (\mu, \sigma)$. The likelihood (dropping constants) is

$$\begin{aligned}\mathcal{L}_n(\mu, \sigma) &= \prod_{i=1}^n \frac{1}{\sigma} \exp \left\{ -\frac{1}{2\sigma^2} (X_i - \mu)^2 \right\} \\ &= \sigma^{-n} \exp \left\{ -\frac{1}{2\sigma^2} \sum_i (X_i - \mu)^2 \right\} \\ &= \sigma^{-n} \exp \left\{ -\frac{nS^2}{2\sigma^2} \right\} \exp \left\{ -\frac{n(\bar{X} - \mu)^2}{2\sigma^2} \right\},\end{aligned}$$

where $\bar{X} = n^{-1} \sum_i X_i$ and $S^2 = n^{-1} \sum_i (X_i - \bar{X})^2$.

The last equality follows from the identity $\sum_i (X_i - \mu)^2 = nS^2 + n(\bar{X} - \mu)^2$, obtained by writing $\sum_i (X_i - \bar{X} + \bar{X} - \mu)^2$ and expanding the square.

Example — Normal (Part 2)

Taking logs,

$$\ell(\mu, \sigma) = -n \log \sigma - \frac{nS^2}{2\sigma^2} - \frac{n(\bar{X} - \mu)^2}{2\sigma^2}.$$

Solving the score equations

$$\frac{\partial \ell(\mu, \sigma)}{\partial \mu} = 0 \quad \text{and} \quad \frac{\partial \ell(\mu, \sigma)}{\partial \sigma} = 0$$

gives

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma} = S.$$

These can be verified to be global maxima of the likelihood.

Is the MLE Unbiased?

Definition — Bias

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta,$$

and $\hat{\theta}_n$ is **unbiased** if $\text{bias}(\hat{\theta}_n) = 0$ for every n .

For the Normal model we found

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Since $\mathbb{E}[\bar{X}] = \mu$, the MLE $\hat{\mu}$ is unbiased. The MLE $\hat{\sigma}^2$ is **not**.

Claim

If $X_1, \dots, X_n$ are IID with $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}(X_i) = \sigma^2$, then

$$\mathbb{E}[S^2] = \frac{n-1}{n} \sigma^2.$$

$$\begin{aligned}\mathbb{E}[S^2] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right] \\&= \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left[\left((X_i - \mu) - (\bar{X} - \mu)\right)^2\right] \\&= \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left[(X_i - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu) + (\bar{X} - \mu)^2\right] \\&= \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left[(X_i - \mu)^2\right] - \frac{2}{n} \mathbb{E}\left[(\bar{X} - \mu) \sum_{i=1}^n (X_i - \mu)\right] + \mathbb{E}[(\bar{X} - \mu)^2]\end{aligned}$$

Proof (2/3)

$$\begin{aligned}\mathbb{E}[S^2] &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - \frac{2}{n} \mathbb{E}[(\bar{X} - \mu) \cdot n(\bar{X} - \mu)] + \mathbb{E}[(\bar{X} - \mu)^2] \\&= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - 2 \mathbb{E}[(\bar{X} - \mu)^2] + \mathbb{E}[(\bar{X} - \mu)^2] \\&= \frac{1}{n} \sum_{i=1}^n \mathbb{V}(X_i) - \mathbb{E}[(\bar{X} - \mu)^2] \\&= \frac{1}{n} (n\sigma^2) - \mathbb{E}[(\bar{X} - \mu)^2] \\&= \sigma^2 - \mathbb{E}[(\bar{X} - \mu)^2]\end{aligned}$$

Proof (3/3)

$$\begin{aligned}\mathbb{E}[(\bar{X} - \mu)^2] &= \mathbb{V}(\bar{X}) = \mathbb{V}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \mathbb{V}\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i) = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}\end{aligned}$$

$$\mathbb{E}[S^2] = \sigma^2 - \frac{\sigma^2}{n} = \left(1 - \frac{1}{n}\right) \sigma^2 = \frac{n-1}{n} \sigma^2$$

$$\boxed{\mathbb{E}[S^2] = \frac{n-1}{n} \sigma^2 \neq \sigma^2} \quad \text{bias}(S^2) = -\frac{\sigma^2}{n}$$

□

Fixing the Bias — Bessel's Correction

Define instead

$$\hat{\sigma}_{\text{unb}}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n}{n-1} S^2.$$

Then

$$\mathbb{E}[\hat{\sigma}_{\text{unb}}^2] = \frac{n}{n-1} \mathbb{E}[S^2] = \frac{n}{n-1} \cdot \frac{n-1}{n} \sigma^2 = \sigma^2.$$

Conclusion

$$\boxed{\mathbb{E}[\hat{\sigma}_{\text{unb}}^2] = \sigma^2}$$

Dividing by $n - 1$ instead of n gives an unbiased estimator. This is known as **Bessel's correction**.

Note that this corrected estimator is *not* the MLE.

What This Tells Us About MLEs

Unbiasedness is not on the list

Maximum likelihood makes the observed data as probable as possible — it makes no promise that $\mathbb{E}[\hat{\theta}_n] = \theta$ for finite n .

- The bias here is $-\sigma^2/n$, which $\rightarrow 0$ as $n \rightarrow \infty$: the MLE is **asymptotically unbiased**, and indeed consistent, even though it is biased at any fixed n .
- Bias is also **not preserved under transformation**: even when $\hat{\theta}_n$ is unbiased for θ , $g(\hat{\theta}_n)$ is generally biased for $g(\theta)$ (unless g is affine), since $\mathbb{E}[g(\hat{\theta}_n)] \neq g(\mathbb{E}[\hat{\theta}_n])$ in general. The MLE, by contrast, *is* equivariant.
- Unbiasedness is not always desirable on its own: what usually matters is the mean squared error,

$$\text{MSE}(\hat{\theta}_n) = \text{bias}(\hat{\theta}_n)^2 + \mathbb{V}(\hat{\theta}_n),$$

and a small bias is often worth a large reduction in variance.

Example — A Hard Example: Intuition

Let $X_1, \dots, X_n \sim \text{Uniform}(0, \theta)$. What is $\hat{\theta}_n$?

- If $\theta < X_i$ for some i , the data would be **impossible** to generate under that θ . So θ must be at least as large as every X_i — that is, $\theta \geq X_{(n)} = \max\{X_1, \dots, X_n\}$.
- But if we make θ **too large**, the density $f(x; \theta) = 1/\theta$ shrinks for every observation, making the data less probable overall.

The sweet spot

The data has the highest probability — while still being possible to generate — exactly at $\theta = X_{(n)}$.

Example — A Hard Example: Formal Proof

Recall $f(x; \theta) = 1/\theta$ for $0 \leq x \leq \theta$, and 0 otherwise.

Fix θ . If $\theta < X_i$ for some i , then $f(X_i; \theta) = 0$, so $\mathcal{L}_n(\theta) = \prod_i f(X_i; \theta) = 0$. Hence $\mathcal{L}_n(\theta) = 0$ whenever $\theta < X_{(n)}$.

Now consider $\theta \geq X_{(n)}$. Then $f(X_i; \theta) = 1/\theta$ for every i , so $\mathcal{L}_n(\theta) = \theta^{-n}$.

Conclusion

$$\mathcal{L}_n(\theta) = \begin{cases} \left(\frac{1}{\theta}\right)^n & \theta \geq X_{(n)} \\ 0 & \theta < X_{(n)}. \end{cases}$$

$\mathcal{L}_n(\theta)$ is strictly decreasing on $[X_{(n)}, \infty)$. Hence

$$\hat{\theta}_n = X_{(n)}. \quad \square$$

Properties of Maximum Likelihood Estimators

Under certain regularity conditions on the model, the MLE $\hat{\theta}_n$ possesses many properties that make it an appealing choice of estimator:

1. The MLE is **consistent**: $\hat{\theta}_n \xrightarrow{P} \theta_*$, where θ_* denotes the true value of θ .
2. The MLE is **equivariant**: if $\hat{\theta}_n$ is the MLE of θ , then $g(\hat{\theta}_n)$ is the MLE of $g(\theta)$.
3. The MLE is **asymptotically Normal**: The distribution of $\hat{\theta}$ approaches a normal distribution as $n \rightarrow \infty$, with a mean of θ and a standard deviation of $\hat{\sigma}$.
4. The MLE is **asymptotically optimal (efficient)**: among all well-behaved estimators, the MLE has the smallest variance, at least for large samples.
5. The MLE is **approximately the Bayes estimator**.

We now look closely at properties 1 and 2: **consistency** and **equivariance**.

Consistency — Setup

Suppose we observe IID data

$$X_1, X_2, \dots, X_n \sim f(x; \theta_0),$$

where θ_0 is the **true** parameter. The MLE is

$$\hat{\theta}_n = \arg \max_{\theta} \prod_{i=1}^n f(X_i; \theta).$$

Because products are inconvenient, we take logs — this does not change the maximizer:

$$\hat{\theta}_n = \arg \max_{\theta} \sum_{i=1}^n \log f(X_i; \theta).$$

Dividing by n *also* does not change the maximizer:

$$\hat{\theta}_n = \arg \max_{\theta} \frac{1}{n} \sum_{i=1}^n \log f(X_i; \theta).$$

Consistency — The Normalized Log-Likelihood

Define the **normalized log-likelihood**

$$L_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(X_i; \theta).$$

For any *fixed* θ , the terms $\log f(X_1; \theta), \dots, \log f(X_n; \theta)$ are IID random variables.

By the Law of Large Numbers

$$L_n(\theta) \xrightarrow{P} \mathbb{E}_{\theta_0} [\log f(X; \theta)] =: L(\theta).$$

So, roughly speaking,

$$\boxed{L_n(\theta) \longrightarrow L(\theta)} \quad \text{as } n \rightarrow \infty.$$

This is the fundamental reason MLEs tend to be consistent.

Consistency — Where Is $L(\theta)$ Maximized?

We claim $L(\theta)$ is maximized at the **true parameter** θ_0 . Consider

$$\begin{aligned} L(\theta_0) - L(\theta) &= \mathbb{E}_{\theta_0} [\log f(X; \theta_0) - \log f(X; \theta)] \\ &= \mathbb{E}_{\theta_0} \left[\log \frac{f(X; \theta_0)}{f(X; \theta)} \right] = D_{\text{KL}}(f_{\theta_0} \parallel f_{\theta}), \end{aligned}$$

the **Kullback–Leibler divergence** between f_{θ_0} and f_{θ} .

Key property of KL divergence

$$D_{\text{KL}}(f_{\theta_0} \parallel f_{\theta}) \geq 0.$$

Therefore $L(\theta_0) - L(\theta) \geq 0$, so

$$\boxed{L(\theta_0) \geq L(\theta)} \quad \text{for every } \theta.$$

Consistency — Uniqueness via Identifiability

If the model is **identifiable**, then

$$f(x; \theta) = f(x; \theta_0) \implies \theta = \theta_0,$$

and consequently

$$L(\theta_0) > L(\theta) \quad \text{for every } \theta \neq \theta_0.$$

Conclusion

The population (limiting) log-likelihood $L(\theta)$ has a **unique maximum** at the true parameter θ_0 .

Consistency — Putting the Pieces Together

Finite sample

$$L_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(X_i; \theta),$$
$$\hat{\theta}_n = \arg \max_{\theta} L_n(\theta).$$

Infinite sample

$$L_n(\theta) \rightarrow L(\theta) = \mathbb{E}_{\theta_0}[\log f(X; \theta)],$$

with unique maximum at θ_0 .

As n increases, $L_n(\theta)$ increasingly resembles a function whose unique maximum is at θ_0 . Hence its maximizer must move toward θ_0 :

$$\hat{\theta}_n \xrightarrow{P} \theta_0.$$

That's **consistency**.

Consistency — Intuition

Think of $L_n(\theta)$ as the log-likelihood curve computed from your *particular* sample. As you average more and more observations, the random fluctuations average out:

sample average $\longrightarrow$ population average

by the law of large numbers.

And the population average has a special property:

$\mathbb{E}_{\theta_0}[\log f(X; \theta)]$ is maximized at θ_0 .

Therefore:

MLE $\longrightarrow$ maximizer of expected log-likelihood $\longrightarrow \theta_0$.

Consistency — A Subtle Issue

The LLN gives $L_n(\theta) \rightarrow L(\theta)$ for a **fixed** θ . But the MLE itself changes with n : $\hat{\theta}_n = \arg \max_{\theta} L_n(\theta)$.

Pointwise convergence of $L_n(\theta) \rightarrow L(\theta)$ for every fixed θ is **not by itself enough** to guarantee that the maximizers converge. We typically also need **uniform convergence**:

$$\sup_{\theta \in \Theta} |L_n(\theta) - L(\theta)| \xrightarrow{P} 0,$$

together with conditions ensuring $L(\theta)$ has a unique maximum at θ_0 .

Then the **argmax theorem** gives

$$\arg \max_{\theta} L_n(\theta) \xrightarrow{P} \arg \max_{\theta} L(\theta) = \theta_0.$$

This is the rigorous mathematical backbone of the MLE consistency proof.

Theorem

Let $\tau = g(\theta)$ be a function of θ . Let $\hat{\theta}_n$ be the MLE of θ . Then

$$\hat{\tau}_n = g(\hat{\theta}_n)$$

is the MLE of τ .

In words: the MLE of a transformed parameter is just the transformation applied to the MLE. We do not need to redo the maximization from scratch.

Equivariance — Proof: For Invertible g

Since g is invertible, $\theta = g^{-1}(\tau)$ is a bijection and $\hat{\theta}_n = g^{-1}(\hat{\tau}_n)$.

The likelihood of τ

Reparametrizing the model by τ gives density $f(x; g^{-1}(\tau))$, so the likelihood of τ is the function

$$\mathcal{L}_n^\tau(\tau) := \prod_{i=1}^n f(x_i; g^{-1}(\tau)) = \mathcal{L}_n(g^{-1}(\tau)),$$

obtained by composing $\mathcal{L}_n$ with g^{-1} .

Let g be a bijection, $\mathcal{L}_n^\tau(\tau) = \mathcal{L}_n(g^{-1}(\tau))$ is largest exactly when $g^{-1}(\tau)$ equals the maximizer $\hat{\theta}_n$ of $\mathcal{L}_n$, i.e. when $\tau = g(\hat{\theta}_n)$. Hence

$$\arg \max_{\tau} \mathcal{L}_n^\tau(\tau) = g(\hat{\theta}_n) = \hat{\tau}_n,$$

so $\hat{\tau}_n$ maximizes the likelihood of τ — that is, $\hat{\tau}_n$ is the MLE of τ . $\square$